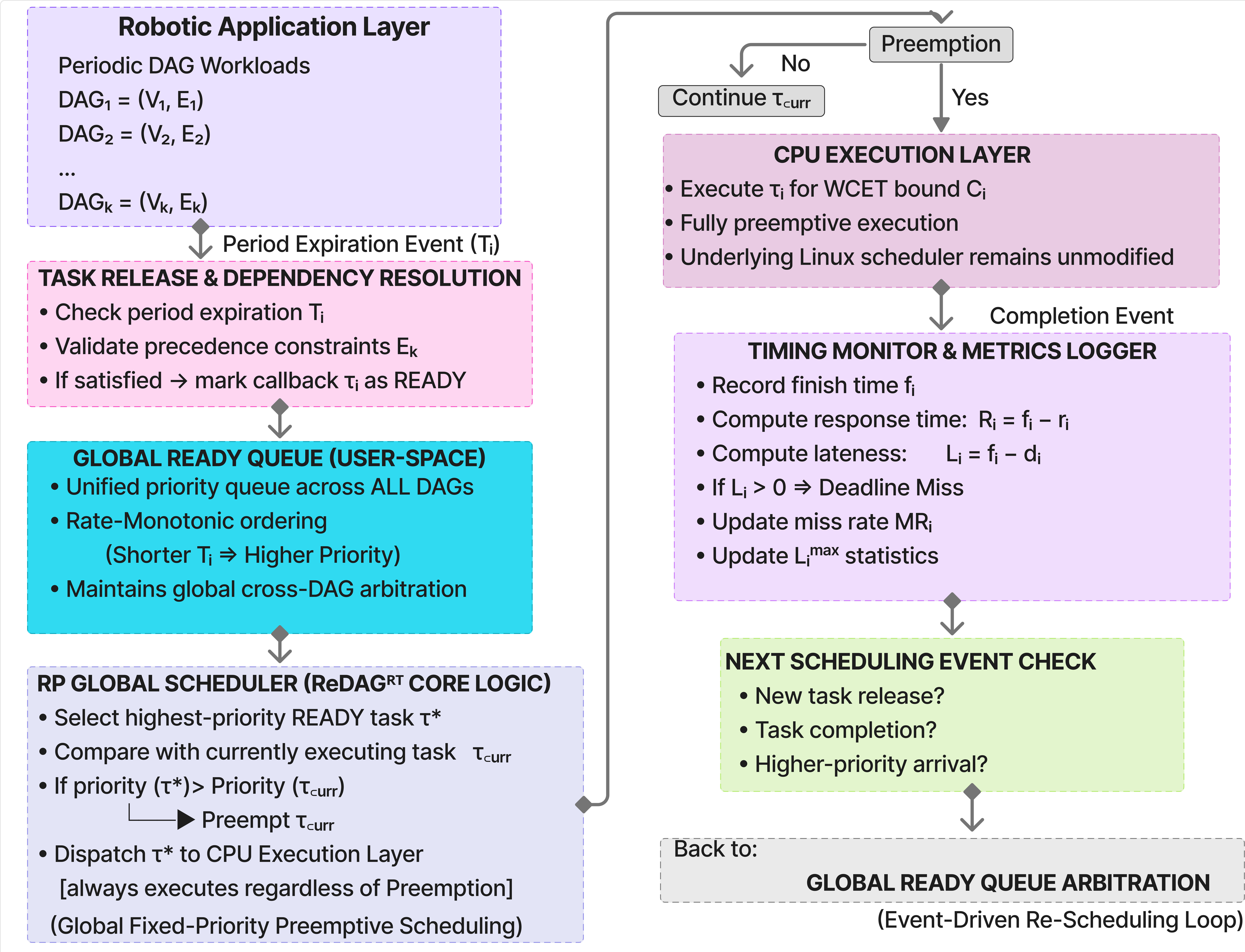
Robotic Application Layer
Periodic DAG Workloads
$DAG_1 = (V_1, E_1)$
$DAG_2 = (V_2, E_2)$
...
$DAG_k = (V_k, E_k)$
Period Expiration Event (T_i)
TASK RELEASE & DEPENDENCY RESOLUTION
• Check period expiration T_i
• Validate precedence constraints E_k
• If satisfied → mark callback τ_i as READY
GLOBAL READY QUEUE (USER-SPACE)
• Unified priority queue across ALL DAGs
• Rate-Monotonic ordering
(Shorter T_i ⇒ Higher Priority)
• Maintains global cross-DAG arbitration
RP GLOBAL SCHEDULER (ReDAG^{RT} CORE LOGIC)
• Select highest-priority READY task τ^*
• Compare with currently executing task τ_{curr}
• If priority (τ^*)> Priority (τ_{curr})
Preempt τ_{curr}
• Dispatch τ^* to CPU Execution Layer
[always executes regardless of Preemption]
(Global Fixed-Priority Preemptive Scheduling)
Preemption
No
Continue τ_{curr}
Yes
CPU EXECUTION LAYER
• Execute τ_i for WCET bound C_i
• Fully preemptive execution
• Underlying Linux scheduler remains unmodified
Completion Event
TIMING MONITOR & METRICS LOGGER
• Record finish time f_i
• Compute response time: $R_i = f_i - r_i$
• Compute lateness: $L_i = f_i - d_i$
• If $L_i > 0$ ⇒ Deadline Miss
• Update miss rate MR_i
• Update L_i^{max} statistics
NEXT SCHEDULING EVENT CHECK
• New task release?
• Task completion?
• Higher-priority arrival?
Back to:
GLOBAL READY QUEUE ARBITRATION
(Event-Driven Re-Scheduling Loop)